Inference & Proofs

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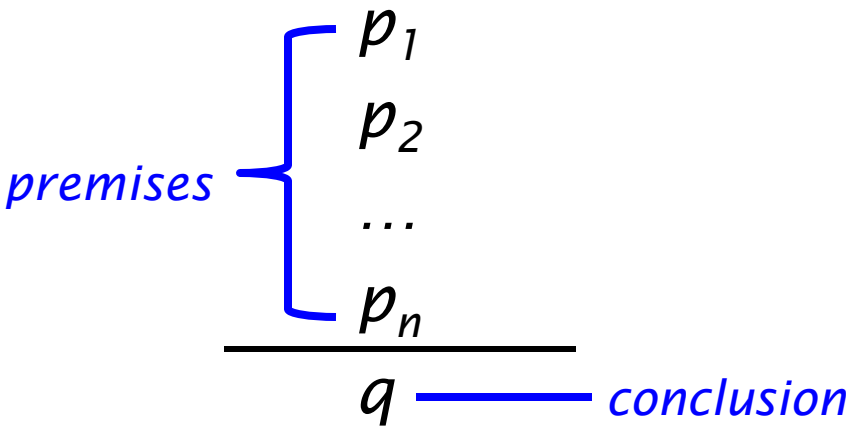
  Kenneth Rosen: Section 1.6, 1.8

Argument

- **Definition**

  An **argument** in propositional logic is a sequence of propositions.

  All but the final proposition in the argument are called **premises** or **hypotheses** and the final proposition is called the **conclusion**.



Argument and Validity

- An argument is **valid** if the truth of all its premises implies that the conclusion is true, independently of which particular propositions are substituted for particular variables. Equivalently, an argument with premises $p_1, p_2, \ldots, p_n$ and conclusion q is **valid** when

$$(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \rightarrow q$$

  is a **tautology**, i.e., $[(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \rightarrow q] \equiv \text{True}$
- Using a truth table to show the validity of an argument can be very tedious
  - if the argument involves n propositional variables, the truth table has 2^n rows.
- This process can be simplified significantly by using rules of inference. Rules of inference are valid argument forms, independently of the underlying propositions.

Rules of Inference for Propositional Logic

Rules of Inference for Propositional Logic		
Rule of inference	Tautology	Name
p $p \to q$ $\therefore \quad q$	$[p \wedge (p \to q)] \to q$	Modus ponens
$\neg q$ $p \to q$ $\therefore \quad \neg p$	$[\neg q \wedge (p \to q)] \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore \quad p \to r$	$[(p \to q) \wedge (q \to r)] \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore \quad q$	$[(p \vee q) \wedge \neg p] \to q$	Disjunctive syllogism

Rules of Inference for Propositional Logic (cont'd)

Rules of Inference for Propositional Logic		
Rule of inference	Tautology	Name
p $\therefore\ p \vee q$	$p \rightarrow (p \vee q)$	Addition
$p \wedge q$ $\therefore\ p$	$(p \wedge q) \rightarrow p$	Simplification
p q $\therefore\ p \wedge q$	$[(p) \wedge (q)] \rightarrow (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore\ q \vee r$	$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee r)$	Resolution

Modus Ponens: The Most Basic Rule

- If p is true and $p \rightarrow q$ is true, then q must be also true.
- Premises: p, $p \rightarrow q$
- Conclusion: q
- Tautology: $[(p \wedge (p \rightarrow q)] \rightarrow q$
- Proof that $[(p \wedge (p \rightarrow q)] \rightarrow q \equiv$ True (independently of p and q)

$[(p \wedge (p \rightarrow q)] \rightarrow q \equiv$ *implication equivalence*

$\neg[(p \wedge (\neg p \vee q)] \vee q \equiv$ *De Morgan*

$[\neg p \vee \neg(\neg p \vee q)] \vee q \equiv$ *De Morgan*

$[\neg p \vee (p \wedge \neg q)] \vee q \equiv$ *Distibutive*

$[(\neg p \vee p) \wedge (\neg p \vee \neg q)] \vee q \equiv$ *Negation Law* $(\neg p \vee p)$ is T

$[(T \wedge (\neg p \vee \neg q)] \vee q \equiv$ *Identity Law* $(T \wedge (\neg p \vee \neg q))$ is $(\neg p \vee \neg q)$

$(\neg p \vee \neg q) \vee q \equiv$ *Associative*

$\neg p \vee (\neg q \vee q) \equiv$ *Negation Law* $(\neg q \vee q)$ is T

$\neg p \vee T \equiv T$ *Identity Law*

Exercise Show the Validity of Modus Tollens

$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$

I must show that this is a tautology; i.e., it is always true

$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p \equiv$

$[\neg q \wedge (\neg p \vee q)] \rightarrow \neg p \equiv$

$[(\neg q \wedge \neg p) \vee (\neg q \wedge q)] \rightarrow \neg p \equiv$

$(\neg q \wedge \neg p) \rightarrow \neg p \equiv$

$\neg(\neg q \wedge \neg p) \vee \neg p \equiv$

$q \vee p \vee \neg p \equiv$

$q \vee T \equiv$ True

Exercise: Show the validity of Disjunctive Syllogism

- $[(p \vee q) \wedge \neg p] \rightarrow q$
- Equivalently show that $[(p \vee q) \wedge \neg p] \rightarrow q \equiv \mathsf{T}$

$[(p \vee q) \wedge \neg p] \rightarrow q \equiv$

$\neg[(p \vee q) \wedge \neg p] \vee q \equiv$

$[\neg(p \vee q) \vee p] \vee q \equiv$

$[(\neg p \wedge \neg q) \vee p] \vee q \equiv$

$[(\neg p \vee p) \wedge (\neg q \vee p)] \vee q \equiv$

$(\neg q \vee p) \vee q \equiv$

$(\neg q \vee q) \vee p \equiv \mathsf{T} \vee p \equiv \mathsf{T}$

Exercise: Show the validity of Resolution

$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee r)$

I must show that this is a tautology

$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow (q \vee r) \equiv$

$\neg[(p \vee q) \wedge (\neg p \vee r)] \vee (q \vee r) \equiv$

$[\neg(p \vee q) \vee \neg(\neg p \vee r)] \vee (q \vee r) \equiv$

$(\neg p \wedge \neg q) \vee (p \wedge \neg r) \vee q \vee r \equiv$

$(\neg p \wedge \neg q) \vee [(p \wedge \neg r) \vee r] \vee q \equiv$

$(\neg p \wedge \neg q) \vee [(p \vee r) \wedge (r \vee \neg r)] \vee q \equiv$

$(\neg p \wedge \neg q) \vee (p \vee r \vee q) \equiv$

$[(\neg p \wedge \neg q) \vee p] \vee r \vee q \equiv$

$[(\neg p \vee p) \wedge (\neg q \vee p)] \vee r \vee q \equiv$

$\neg q \vee p \vee r \vee q \equiv$

$\neg q \vee q \vee p \vee r \equiv$

$T \vee p \vee r \equiv$ True

Example on Application of Resolution rule

- Given the hypotheses "it is not snowing or Jasmine is skiing" and "it is snowing or Bart is playing hockey" draw a conclusion.
- *s*: it is snowing
- *k*: Jasmine is skiing
- *h*: Bart is playing hockey
- **Solution**

$$\begin{array}{l} \neg s \vee k \\ \quad s \vee h \\ \hline \therefore\ k \vee h \end{array} \quad \text{(by resolution)}$$

"Jasmine is skiing or Bart is playing hockey"

Example on application of Resolution, Simplification rules

- Show that the hypotheses $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.
- **Solution**

1. $(p \wedge q) \vee r$	Premise
2. $r \rightarrow s$	Premise
3. $\neg r \vee s$	2, equivalence
4. $(p \wedge q) \vee s$	1,3 resolution
5. $(p \vee s) \wedge (q \vee s)$	4, equivalence
6. $p \vee s$	5, simplification

Example on Application of multiple rules of inference

- We will use the following propositions
  - p: "it is sunny this afternoon."
  - q: "it is colder than yesterday."
  - r: "we will go swimming."
  - s: "we will take a canoe trip."
  - t: "we will be home by sunset"
- Express the following statements in propositional logic
  - "it is not sunny this afternoon and it is colder than yesterday"
  - $\neg p \wedge q$
  - "we will go swimming only if it is sunny"
  - $r \rightarrow p$
  - "if we do not go swimming, then we will take a canoe trip"
  - $\neg\, r \rightarrow s$
  - "if we take a canoe trip, then we will be home by sunset"
  - $s \rightarrow t$

Example on Application of multiple rules of inference 2

- Given premises: $\neg p \wedge q$, $r \rightarrow p$, $\neg r \rightarrow s$, and $s \rightarrow t$, conclude t

1 $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Example 1 of **Fallacy** (invalid argument)

- Is the argument valid:

"if you were a good student, you are rich".

"You are rich"

Therefore: "You were a good student"

- p: "You were a good student "
- q: "You are rich"

$$\begin{array}{l} p \rightarrow q \\ q \\ \hline p \end{array}$$

- Wrong because $p \rightarrow q$ is not equivalent to $q \rightarrow p$. p = F, q = T makes the premises true, but the conclusion false.

Example 2 of **Fallacy** (invalid argument)

- Example: Is the following argument valid?
  - Premises:
  - $p \rightarrow r$
  - $q \rightarrow r$
  - $\neg (p \vee q)$
  - Conclusion: $\neg r$
- Solution: The argument is invalid.
  - The truth assignment p = F, q = F, r =T makes all the premises true, but the conclusion false.

Rules of Inference for Predicate Logic

Rules of Inference for Predicate Logic	
Rule of inference	Name
$\forall x\, P(x)$ $\therefore\ P(c)$	Universal instantiation
$P(c)$ for an arbitrary c $\therefore\ \forall x\, P(x)$	Universal generalization
$\exists x\, P(x)$ $\therefore\ P(c)$ for some element c	Existential instantiation
$P(c)$ for some element c $\therefore\ \exists x\, P(x)$	Existential generalization

Rules for Propositions and Quantified Statements

- Rules for propositions and quantified statements can be combined.
- Because universal instantiation and modus ponens are used together very often, we call this combination **universal modus ponens**.
- The rule states that if $\forall x\,(P(x) \rightarrow C(x))$ and $P(a)$ is true for a particular element in the domain, then $C(a)$.
- Similarly, **universal modus tollens** states that if $\forall x\,(P(x) \rightarrow C(x))$ and $\neg\, C(a)$ is true for a particular element in the domain, then $\neg\, P(a)$.

Example on (Universal) Modus Ponens

- Show that the premises "all men must die" and "John Snow is a man" imply the conclusion "John Snow must die", using the following predicates.
  - $P(x)$: x is a man
  - $C(x)$: x must die
- **Solution**

1. $\forall x\,(P(x) \rightarrow C(x))$ Premise
2. $P(John\ Snow)$ Premise
3. $P(John\ Snow) \rightarrow C(John\ Snow)$ 1 Universal instantiation
4. $C(John\ Snow)$ 2,3 Modus ponens

Combining 3. and 4. into a single step

1. $\forall x\,(P(x) \rightarrow C(x))$ Premise
2. $P(John\ Snow)$ Premise
3. $C(John\ Snow)$ 1,2 Universal Modus ponens

Example of Multiple Rules

- Show that the premises "a student in this class has not read the textbook" and "everyone in this class passed the course" imply the conclusion "someone who passed the course has not read the textbook".
- Predicates:
  - $P(x)$: x is a student in this class
  - $R(x)$: x has read the textbook
  - $U(x)$: x passed the course
- Premises
- "A student in this class has not read the textbook": $\exists x\,(P(x) \wedge \neg R(x))$
- "Everyone in this class passed the course": $\forall x\,(P(x) \rightarrow U(x))$
- Conclusion
- "Someone who passed the course has not read the textbook": $\exists x\,(U(x) \wedge \neg R(x))$

Example of Multiple Rules (cont)

From Premises: $\exists x\,(P(x) \wedge \neg R(x))$ and $\forall x\,(P(x) \rightarrow U(x))$, draw the conclusion: $\exists x\,(U(x) \wedge \neg R(x))$

1. $\exists x\,(P(x) \wedge \neg R(x))$	Premise
2. $P(a) \wedge \neg R(a)$ for some a	1, Existential instantiation
3. $P(a)$	2, Simplification
4. $\neg R(a)$	2, Simplification
5. $\forall x\,(P(x) \rightarrow U(x))$	Premise
6. $P(a) \rightarrow U(a)$	5, Universal instantiation
7. $U(a)$	3,6, Modus ponens
8. $U(a) \wedge \neg R(a)$	4,7, Conjunction
9. $\exists x\,(U(x) \wedge \neg R(x))$	8, Existential generalization

Previous Exercise from Predicate Logic

- Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a lion", "x is fierce", and "x drinks coffee", respectively. Assume that the domain consists of all creatures.
- Consider the following argument:

1. Premise 1: "All lions are fierce"
2. Premise 2: "Some lions do not drink coffee"
3. Conclusion: "Some fierce creatures do not drink coffee"

- Statements using quantifiers and $P(x)$, $Q(x)$, and $R(x)$.

1. $\forall x\,(P(x) \rightarrow Q(x))$
2. $\exists x\,(P(x) \wedge \neg\, R(x))$
3. $\exists x\,(Q(x) \wedge \neg\, R(x))$

- **Show how to reach the conclusion through inference rules**

Previous Exercise continued 2

1. Premise 1: $\forall x\,(P(x) \rightarrow Q(x))$
2. Premise 2: $\exists x\,(P(x) \wedge \neg R(x))$
3. Conclusion: $\exists x\,(Q(x) \wedge \neg R(x))$

- **Show how to reach the conclusion through inference rules**

1.	$\exists x\,(P(x) \wedge \neg R(x))$	Premise
2.	$P(a) \wedge \neg R(a)$	1. Existential instantiation
3.	$\neg R(a)$	2. Simplification
4.	$P(a)$	2. Simplification
5.	$\forall x\,(P(x) \rightarrow Q(x))$	Premise
6.	$P(a) \rightarrow Q(a)$	5. Universal instantiation
7.	$Q(a)$	4.,6. Modus Ponens
8.	$Q(a) \wedge \neg R(a)$	3.,7. Conjunction
9.	$\exists x\,(Q(x) \wedge \neg R(x))$	8. Existential generalization

The steps that we used are the same as in the previous example. The actual predicates (e.g., meaning of $P(x)$) and the domains are not important.

Terminology on Proof Techniques

- **Definition:** An **axiom** is a statement that is assumed to be true.
- **Definition:** A **theorem** is a statement that can be shown to be true.
- **Definition:** A less important theorem that is helpful in the proof of other theorems is called a **lemma**.
- **Definition:** A **proof** is a valid argument that establishes the truth of a theorem. The statements used in a proof can include axioms, premises of the theorem, and previously proved theorems or lemmas. Using these ingredients and rules of inference, the final step of the proof establishes the truth of the statement being proved.

More Terminology

- **Definition**

  A **corollary** is a theorem that can be established directly from a theorem that has been proved.

- **Definition**

  A **conjecture** is a statement that is being proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert.

- **Remark**

  When a proof of a conjecture is found, the conjecture becomes a theorem. However, many conjectures are eventually found to be false.

Direct Proof

- A **direct proof** of a conditional statement $p \rightarrow q$

  The first step is the assumption that p is true.

  Subsequent steps are constructed using axioms, definitions, previously proved theorems, and rules of inference, with the final step showing that q must also be true.

Example on Direct Proof

- Definition: The integer n is even if there exists an integer k such that $n = 2k$, and n is odd if there exists an integer k such that $n = 2k + 1$.

- Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."
- Equivalent statement: $\forall n\ (P(n) \rightarrow Q(n))$, where $P(n)$ is "n is an odd integer" and $Q(n)$ is "n^2 is odd", where the domain of n is all integers.
- Assume the hypothesis $P(n)$ is true: $n=2k+1$.
- We must show that the conclusion $Q(n)$ is also true.
- $n^2 =(2k+1)^2= 4k^2+ 4k + 1 = 2(2k^2+ 2k)+1 = 2k'+1$.
- Thus, $n^2= 2k'+1$ for some integer k'; n^2 is odd.

Connection between Direct Proofs and Inference in Predicate Logic

- I wish to prove $\forall n\,(P(n) \rightarrow Q(n))$ (e.g., $\forall n\,(Odd(n) \rightarrow Odd(n^2))$
- I use *universal instantiation* to select an *arbitrary* element c of the domain, such that $P(c)$ is True.
  - e.g., an arbitrary odd integer $c=2k+1$
- I show that $P(c) \rightarrow Q(c)$ is True.
  - e.g., $Odd(c) \rightarrow Odd(c^2)$
- I apply *universal generalization* to derive $\forall n\,(P(n) \rightarrow Q(n))$
- We skip universal instantiation and generalization in our proofs.

Exercise 1 on Direct Proof

- Definition: An integer a is a perfect square if there is an integer b such that $a = b^2$

- Give a direct proof of: "if m and n are both perfect squares, then mn is also a perfect square"
- Assume the hypothesis is true: $m=s^2$ and $n=t^2$, where s,t are integers.
- We must show that mn is a perfect square.
- $mn = s^2t^2 = (st)^2$. Thus, mn is a perfect square (of $s \cdot t$).

Exercise 2 on Direct Proof

- Definition: A real number r is rational if there exist integers p and q with $q \neq 0$ such that $r = p/q$. A real number that is not rational is called irrational

- Give a direct proof for: "the sum of two rational numbers is rational." Equivalently, "for every real number r and every real number s, if r and s are rational numbers, then $r + s$ is rational.
- Assume the hypothesis is true: $r = p/q$ and $s = t/u$, with $q \neq 0$ and $u \neq 0$.
- $r+s = p/q + t/u = (pu + tq)/qu$.
- Since $q \neq 0$ and $u \neq 0$, we express $r+s$ as the ratio of two integers $(pu + tq)$ and qu. Thus, $r+s$ is rational.

Limitation of Direct Proofs

- Direct proofs are useful but attempts at direct proofs sometimes lead to dead ends. There are other proof techniques.

Proofs that are not direct proofs, i.e., they do not start with the hypothesis and end with the conclusion, are called **indirect proofs**.

We will consider several types of indirect proofs.

Proof by Contraposition

- A **proof by contraposition** makes use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contrapositive $\neg q \rightarrow \neg p$.

- This means that the conditional statement $p \rightarrow q$ can be proved by showing that its contrapositive $\neg q \rightarrow \neg p$ is true.

- To do so, we take $\neg q$ as a hypothesis, and using axioms, definitions, previously proved theorems, and rules of inference, we show that $\neg p$ must follow.

Example Proof by Contraposition

- Give a direct proof of "If n is an integer, and $3n+2$ is odd, then n is odd."
- $P(n)$: $3n+2$ is odd, $Q(n)$: n is odd
- Direct proof: $P(n) \rightarrow Q(n)$
- Assume that $3n+2$ is odd. Then, $3n+2= 2k+1$ for some integer k. Thus, $n=(2k-1)/3$ … stuck.
- Proof by Contraposition: $\neg Q(n) \rightarrow \neg P(n)$
- Assume that the conclusion is false, i.e., n is even: $n=2k$. Then I will prove that the premise is also false, i.e., $3n+2$ must be even.
- $3n+2 = 3(2k)+2 = 6k+2 = 2(3k+1)=2k'$ *n even*

Example Proof by Contraposition 2

- Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.
- $P(n)$: $n = ab$, $Q(n)$: $(a \leq \sqrt{n} \vee b \leq \sqrt{n})$
- Direct proof: $P(n) \rightarrow Q(n)$
- Proof by Contraposition: $\neg Q(n) \rightarrow \neg P(n)$
- $\neg Q(n)$: $a > \sqrt{n} \wedge b > \sqrt{n}$.
- I will prove that $\neg P(n)$, that is, $n \neq ab$.
- $ab > \sqrt{n}\sqrt{n} = n$.

Exercise on Proof by Contraposition

- Prove that if n is an integer and n^2 is even, then n is even.
- $P(n)$: n^2 is even, $Q(n)$: n is even
- Proof by Contraposition: $\neg Q(n) \rightarrow \neg P(n)$
- Assume $\neg Q(n)$, i.e., n is odd. Then I will prove $\neg P(n)$, i.e., n^2 must be odd.
- $n = 2k+1 \Rightarrow n^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1 = 2k'+1$ *n^2 odd*

Proof by Contradiction

- Suppose we want to prove that a statement p is true.

- Instead, we assume p is false, i.e., $\neg p$ is true. Then, using axioms, definitions, previously proved theorems, and rules of inference, we derive a contradiction **F.** This means that our assumption that "$\neg p$ is true" is false. Consequently, p must be true.

- If you want to prove p, you show $\neg p \rightarrow \mathbf{F}$
- An example of contradiction is $r \wedge \neg r$, where r is a proposition, .i.e., $\neg p \rightarrow r \wedge \neg r$.

Example Proof by Contradiction

- Prove that $\sqrt{2}$ is irrational.
- Assume that $\sqrt{2}$ is rational. We will show that this leads to a contradiction.
- There exist integers a and b with $\sqrt{2} = a/b$, where a and b have no common factors so that the fraction a/b is in lowest terms (e.g., instead of 4/12, we will use 1/3).
- $\sqrt{2} = a/b \Rightarrow 2=(a^2/b^2) \Rightarrow 2b^2 = a^2$.
- Therefore, a^2 is even. As we have shown in the exercise for contraposition, a is also even. Thus, there exists integer c such that $a=2c$.
- $2b^2 = a^2 \Rightarrow 2b^2 = (2c)^2 \Rightarrow 2b^2 = 4c^2 \Rightarrow b^2 = 2c^2$
- Thus, b^2 is even, which implies that b is even.
- 2 divides a and b, which contradicts the assumption that the fraction a/b is in lowest terms.

Proof by Contradiction for Conditional Statement

- Proofs by contradiction can be used to prove conditional statements $p \rightarrow q$.

We first assume that the negation of the conclusion is true. We then use the premises of the theorem in conjunction with the negation of the conclusion to arrive at a contradiction.

That is, $(p \wedge \neg q) \rightarrow \mathbf{F}$.

The validity of such proofs is based on the logical equivalence of $p \rightarrow q$ and $(p \wedge \neg q) \rightarrow \mathbf{F}$.

Proof by Contradiction for Conditional Statement (cont'd)

- We can rewrite a proof by contraposition of a conditional statement $p \rightarrow q$ as a proof by contradiction.
- In a proof by contraposition, we assume that $\neg q$ is true and then show that $\neg p$ must also be true.
- To rewrite as a proof by contradiction, we suppose that both p and $\neg q$ are true. Then, we use the steps from the proof of $\neg q \rightarrow \neg p$ to show that $\neg p$ is true. This leads to the contradiction $p \wedge \neg p$, completing the proof.

Example of rewriting a proof by contraposition of a conditional statement $p \rightarrow q$ as a proof by contradiction

- "If integer $3n+2$ is odd, then n is odd."
- P: $Odd(3n+2)$ Q: $Odd(n)$, $P \rightarrow Q$
- Proof by contraposition: $\neg Q \rightarrow \neg P$
- Assume $\neg Q$, i.e., even $n=2k$. Then, I will prove $\neg P$, i.e., even $3n+2$
- $3n+2 = 3(2k)+2 = 6k+2 = 2(3k+1)=2k'$
- Proof by contradiction:
- Assume $\neg Q$, i.e., even $n=2k$ and P, i.e., odd $3n+2$. Then, I will reach a contradiction.
- $3n+2=3(2k)+2=6k+2=2(3k+1)=2k'$. $3n+2$ is even, which contradicts the assumption that $3n+2$ is odd.

Exercise On Proof by Contraposition and Contradiction

A real number is *rational* if and only if it can be written as a/b, where a and b are integers. Let x and y be two real numbers such that $xy \neq 0$.

(a) Show, by contraposition, that if *x is irrational*, then $x^{1/2}$ is irrational ($x^{1/2}$ is the square root of x).

Assume that $x^{1/2}$ is rational, that is $x^{1/2} = a/b$, where a and b are integers. Then $x = x^{1/2} \cdot x^{1/2} = a^2/b^2$, where a^2 and b^2 are also integers. Thus, x is rational

(b) Show, by contradiction, that if x is irrational and xy is rational, then y is irrational.

Assume that the premise is true: x is irrational and xy is rational

and the conclusion is false: y is rational. This should lead to a contradiction.

Since y is rational, $y = a/b$ and $1/y = b/a$

Since xy is rational $xy = c/d$, *where a,b,c,d are all integers.*

Then $x = (xy)1/y = cb/ad$, which contradicts that x is irrational

Proving Biconditional Statements

- To prove a biconditional statement of the form $p \leftrightarrow q$, we show that both $p \rightarrow q$ and $q \rightarrow p$ are true.
  - Based on the tautology $(q \leftrightarrow p) \leftrightarrow ((p \rightarrow q) \wedge (q \rightarrow p))$.
- To prove that propositions p_1, p_2, p_3,..., p_n are equivalent ($p_1 \leftrightarrow p_2 \leftrightarrow p_3 \cdots \leftrightarrow p_n$), we show that all $p_1 \rightarrow p_2$, $p_2 \rightarrow p_3$,..., $p_n \rightarrow p_1$ are true.
  - Based on the tautology $p_1 \leftrightarrow p_2 \leftrightarrow p_3 \cdots \leftrightarrow p_n \leftrightarrow (p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \cdots \wedge (p_n \rightarrow p_1)$..

Exercise on Equivalence of Multiple Propositions

Show that these statements about the integer n are logically equivalent:

p_1: n is even, p_2: n-1 is odd, p_3: n^2 is even

I must show that $p_1 \rightarrow p_2$, $p_2 \rightarrow p_3$ and $p_3 \rightarrow p_1$

- Proof of $p_1 \rightarrow p_2$ (direct)

Suppose that n is even (n=2k). Consequently, $n-1 = 2k-1 = 2(k-1)+1 = 2k'+1$. Thus, $n-1$ is odd.

- Proof of $p_2 \rightarrow p_3$ (direct)

Suppose that $n-1$ is odd ($n-1=2k+1$). Then, $n=2k+2$, and $n^2=(2k+2)^2 = 4k^2+8k+4 = 2(2k^2+4k+2) = 2k'$. Thus, n^2 is even.

- Proof of $p_3 \rightarrow p_1$ (contraposition)
- We prove that if n is not even, then n^2 is not even. See Exercise on Proof by Contraposition.

Mistakes in Proofs

What is wrong with this famous supposed "proof" that 1 = 2?

Let a and b be two equal positive integers.

1. $a = b$	Given
2. $a^2 = ab$	Multiply both sides of (1) by a
3. $a^2 - b^2 = ab - b^2$	Subtract b^2 from both sides of (2)
4. $(a - b)(a + b) = b(a - b)$	Factor both sides of (3)
5. $a + b = b$	Divide both sides of (4) by $a - b$
6. $2b = b$	Replace a by b in (5) because $a = b$
7. $2 = 1$	Divide both sides of (6) by b

Error: Step 5 where we divided both sides by $(a - b)$. The error is that $(a - b)$ equals zero. Division of both sides of an equation by the same quantity is valid as long as this quantity is not zero.

Some comments on proofs

- There are many other proof methods
  - We will go over some later; many will not be covered in this course.
- Constructing proofs is an art that can be learned only by trying various lines of attack. There are no fixed procedures for proving theorems.
- Many statements that appear to be theorems have resisted the persistent efforts of mathematicians for hundreds of years.
- For instance, Goldbach's conjecture: "every even positive integer greater than 2 is the sum of two primes" has not yet been proved, and no counterexample has been found.

Automated Theorem Proving

- Hilbert's program: A computer program that, starting from a finite number of axioms and using the rule of inference, automatically decides the truth of any mathematical statement
- Gödel's incompleteness theorem: This is not possible
- There are several automated theorem proving systems that can prove certain subsets of mathematics
- Recently AI models have proven conjectures and theorems that remained unsolved for many years



